ETALE-LOCAL CHARACTERIZATION OF RELATIVELY PERFECT COMPLEXES ON ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 12338-12364, immediately following the statement at 12312-12336.

Editorial note. Clause (4)'s equivalence is imported from the schemes chapter by citation (proof lines 12365-12367); the reviewer should read that import against the bridge lemmas recorded in standard_prerequisites.

Editorial note. Four cross-chapter pointer families (Derived Categories of Spaces x3, Derived Categories of Schemes x1) are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Small etale site versus scheme underlying an algebraic space. Derived Categories of Spaces, Lemma spaces-perfect-lemma-derived-quasi-coherent-small-etale-site, cited at proof lines 12347-12349. Identifies E|_U in D_QCoh(O_U) with an object E_0 of D_QCoh(O_{U_0}) on the underlying schemes, the bridge on which all four clauses are compared.
- Pseudo-coherence descends across the bridge. Derived Categories of Spaces, Lemma spaces-perfect-lemma-descend-pseudo-coherent, cited at proof lines 12350-12351. E|_U is pseudo-coherent if and only if E_0 is, for clauses (2)/(3) to speak the scheme language.
- Relative tor dimension across the bridge. Derived Categories of Spaces, Lemma spaces-perfect-lemma-tor-dimension-rel, cited at proof lines 12352-12356. Finite tor dimension over g^{-1}O_V transfers to finite tor dimension of E_0 over g_0^{-1}O_{V_0}.
- Affine-local characterization on schemes. Derived Categories of Schemes, Lemma perfect-lemma-affine-locally-rel-perfect, cited at proof lines 12365-12367. The equivalence of clause (4) with clauses (2) and (3) is exactly the scheme-side statement, imported verbatim.
- Etale locality of pseudo-coherence and of finite tor dimension. standard facts invoked at proof lines 12359-12361 (pseudo-coherence etale local; locally finite tor dimension etale local). Makes Y-perfectness testable etale locally, which powers (1) => (2) and (3) => (1).

Source reference. Original reference 0DI0: focused-seeds/geometry/S9-spaces-affine-locally-rel-perfect/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0DI2: focused-seeds/geometry/S9-spaces-affine-locally-rel-perfect/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 071Q: focused-seeds/geometry/S9-spaces-affine-locally-rel-perfect/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08HE: focused-seeds/geometry/S9-spaces-affine-locally-rel-perfect/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0DK7: focused-seeds/geometry/S9-spaces-affine-locally-rel-perfect/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08GH: focused-seeds/geometry/S9-spaces-affine-locally-rel-perfect/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is flat and locally of finite presentation. Let $E \in D_{QCoh}(\mathcal{O}_X)$. The following are equivalent:*

(1) *E is Y-perfect,*
(2) *for every commutative diagram*

$$\begin{array}{ccc} U & \xrightarrow{g} & V \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

where U, V are schemes and the vertical arrows are étale, the complex $E|_U$ is V-perfect in the sense of Derived Categories of Schemes, Definition 0DI0,
(3) *for some commutative diagram as in (2) with $U \to X$ surjective, the complex $E|_U$ is V-perfect in the sense of Derived Categories of Schemes, Definition 0DI0,*
(4) *for every commutative diagram as in (2) with U and V affine the complex $R\Gamma(U, E)$ is $\mathcal{O}_Y(V)$-perfect.*

Proof. To make sense of parts (2), (3), (4) of the lemma, observe that the object $E|_U$ of $D_{QCoh}(\mathcal{O}_U)$ corresponds to an object E_0 of $D_{QCoh}(\mathcal{O}_{U_0})$ where U_0 denotes the scheme underlying U, see Derived Categories of Spaces, Lemma 071Q. Moreover, in this case E_0 is pseudo-coherent if and only if $E|_U$ is pseudo-coherent, see Derived Categories of Spaces, Lemma 08HE. Also, $E|_U$ locally has finite tor dimension over $f^{-1}\mathcal{O}_Y|_U = g^{-1}\mathcal{O}_V$ if and only if E_0 locally has finite tor dimension over $g_0^{-1}\mathcal{O}_{V_0}$ by Derived Categories of Spaces, Lemma 0DK7. Here $g_0 : U_0 \to V_0$ is the morphism of schemes representing $g : U \to V$ (notation as in Derived Categories of Spaces, Remark 08GH). Finally, observe that "being pseudo-coherent" is étale local and of course "having locally finite tor dimension" is étale local. Thus we see that it suffices to check Y-perfectness étale locally and by the above discussion we see that (1) implies (2) and (3) implies (1). Since part (4) is equivalent to (2) and (3) by Derived Categories of Schemes, Lemma 0DI2 the proof is complete. □